

PAPER_277 — UQFF Gravitational Recession Damping Factor $\kappa_{\text{recession}}$ for Positive Redshift

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Abstract

We introduce the UQFF Gravitational Recession Damping Factor $\kappa_{\text{recession}} = 1/(1+z)$ for galaxies receding from the observer (positive cosmological redshift $z > 0$). Applied to the Sombrero Galaxy (M104, $z = +0.0063$), this factor yields $\kappa_{\text{recession}} = 0.99374$, attenuating the total UQFF gravitational output by 0.63% relative to the rest-frame value. This result is the precise complement of PAPER_273's blueshift amplifier (Andromeda, $z = -0.001$, $\kappa > 1$), and together the two whitepapers establish the **Universal UQFF Bidirectional Redshift Law**: the single analytic function $\kappa(z) = 1/(1+z)$ applies for all z , unifying recession and approach within one UQFF correction framework.

1. Motivation and Context

In standard cosmological physics, the gravitational interaction between two mass-energy concentrations is not generally modulated by their relative cosmological line-of-sight velocity. However, within the UQFF (Unified Quantum Field Framework), which models gravity as a buoyancy-mediated emergent phenomenon in the quantum vacuum, the energy-density of the mediating vacuum field is Doppler-shifted by the relative cosmological recession. This introduces a multiplicative correction to the full UQFF g_{total} equation.

The Sombrero Galaxy (M104) is a nearby edge-on Sa-type spiral galaxy at a recession velocity of approximately +1890 km/s, corresponding to a spectroscopic heliocentric redshift of $z = +0.0063$. Its recession — the opposite of Andromeda's blueshift approach — provides the ideal test case for the positive- z branch of the universal $\kappa(z)$ function.

This paper establishes: 1. The analytic form of $\kappa_{\text{recession}}$ for $z > 0$. 2. The complementary relationship with PAPER_273 ($z < 0$, approach, blueshifted). 3. The universal UQFF Bidirectional Redshift Law valid for all real $z > -1$. 4. Cosmological limiting cases: early universe gravity attenuation and merger-limit singularity.

2. Theoretical Derivation

2.1 UQFF Vacuum Field Energy Under Recession

In UQFF, the gravitational term g_{UQFF} is proportional to the local vacuum energy density ρ_{vac} at the point of computation. For a source receding from the observer at cosmological velocity corresponding to redshift z , the energy of each mediating vacuum quantum is red-shifted by the factor $(1+z)$:

$$E_{\text{obs}} = \frac{E_{\text{emit}}}{1+z}$$

Since the gravitational coupling in UQFF is proportional to the vacuum field energy density, the full UQFF g_{total} is correspondingly attenuated:

$$g_{\text{UQFF,obs}} = \frac{g_{\text{UQFF,emit}}}{1+z} = \kappa_{\text{recession}} \cdot g_{\text{UQFF,emit}}$$

Defining the recession damping factor:

$$\kappa_{\text{recession}} = \frac{1}{1+z}$$

For the Sombrero Galaxy:

$$\kappa_{\text{recession}}^{\text{Sombrero}} = \frac{1}{1+0.0063} = \frac{1}{1.0063} = 0.99374$$

2.2 Position in the computeG() Equation

The $\kappa_{\text{recession}}$ factor enters the Sombrero UQFF Master Gravity Equation as an outer multiplier:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{g,\text{sum}}^{(26)} + \Lambda_{\text{term}} + g_{\text{quantum}} + g_{\text{Lorentz}} + g_{\text{fluid}} + F_{\text{ring}}(t) + g_{\text{BH}} + g_{\text{exp}} + a_{\text{dust}} + g_{\text{D}}$$

where $\sigma_{\text{sc}} = 1 - B/B_{\text{crit}}$ (superconductivity correction, UNIQUE to Sombrero — no other UQFF module currently uses a dual outer multiplier).

Gravitational attenuation for Sombrero:

$$\Delta g_{\text{recession}} = g_{\text{UQFF}} \left(1 - \frac{1}{1.0063} \right) = 0.00626 \cdot g_{\text{UQFF}}$$

With $g_{\text{base}} = G \cdot M/r^2 = 2.382 \times 10^{-10} \text{ m/s}^2$ and $\text{pre_sum_Ug} \approx 52 \cdot g_{\text{base}} \approx 1.238 \times 10^{-8} \text{ m/s}^2$:

$$\Delta g \approx 0.00626 \times 1.238 \times 10^{-8} \approx 7.75 \times 10^{-11} \text{ m/s}^2$$

3. Universal UQFF Bidirectional Redshift Law

Combining PAPER_273 ($z = -0.001$, Andromeda) and PAPER_277 ($z = +0.0063$, Sombrero), we arrive at the general statement:

Universal UQFF Bidirectional Redshift Law:

$$\kappa(z) = \frac{1}{1+z}, \quad z \in (-1, +\infty)$$

Regime	z range	$\kappa(z)$	Physical meaning
Approach/Blueshift	$z < 0$	> 1	UQFF gravity amplified (PAPER_273)
Rest frame	$z = 0$	1.0	Unmodified UQFF
Recession/Redshift	$z > 0$	< 1	UQFF gravity damped (PAPER_277)

3.1 $\kappa(z)$ at Representative Redshifts

Redshift z	$\kappa(z)$	Astrophysical Context
$z = -0.001$	1.001001	Andromeda (PAPER_273, approach)
$z = 0$	1.000000	Rest frame / MW local group
$z = +0.0063$	0.99374	Sombrero M104 (PAPER_277)
$z = +0.1$	0.90909	Virgo Cluster periphery
$z = +0.5$	0.66667	Intermediate cosmological
$z = +1.0$	0.50000	Halfway epoch ($t \approx 5.8$ Gyr)
$z = +3.5$	0.22222	Epoch of reionisation
$z \rightarrow \infty$	$\rightarrow 0$	Big Bang limit — gravity switchoff
$z \rightarrow -1$	$\rightarrow \infty$	Singularity — merger/coalescence

3.2 Cosmological Limiting Cases

Early Universe ($z \rightarrow \infty$):

$$\lim_{z \rightarrow \infty} \kappa(z) = 0$$

In the UQFF framework this corresponds to **gravitational switchoff** in the extreme early universe — the vacuum field quanta are so severely redshifted that they carry negligible energy. This provides a natural UQFF mechanism for the observed suppression of large-scale structure formation at very high redshift.

Merger Singularity ($z \rightarrow -1$):

$$\lim_{z \rightarrow -1} \kappa(z) = +\infty$$

For a source approaching the observer at the speed of light ($z \rightarrow -1$ in the blueshift convention), the UQFF gravity diverges. This represents the merger or coalescence singularity, where the UQFF vacuum field collapses to a single point and gravitational focusing becomes unbounded.

4. Module Implementation

```
// PAPER_277 – SOMBRERO_UQFF_MODULE.cpp, updateCache()
kappa_recession = 1.0 / (1.0 + z); // z = +0.0063 → kappa_recession = 0.99374

// Applied in computeG():
double g_total = g_sum * kappa_recession * corr_SC;
//                ↑ PAPER_277      ↑ SC correction (unique dual outer multiply)
```

Computed values for Sombrero M104: - $z = +0.0063$ - $\kappa_{\text{recession}} = 0.99374$ - Gravitational damping: -0.626% of g_{UQFF} (recession)

5. Key Constants Introduced

Symbol	Value	Units	Description
$\kappa_{\text{recession}}$	0.99374	dimensionless	UQFF recession damping factor for $z=+0.0063$
z_{Sombrero}	+0.0063	dimensionless	Heliocentric spectroscopic recession redshift
$\Delta g_{\text{recession}}$	$\sim 7.75 \times 10^{-11}$	m/s^2	Absolute gravitational attenuation

6. Physical Significance

1. **Bidirectional completeness:** PAPER_273 and PAPER_277 together show that the UQFF $\kappa(z)$ correction is a universal single-parameter function. No new free parameters are introduced.
2. **Observable consequence for Sombrero:** The 0.626% attenuation is below current measurement precision for M104's rotation curves ($\sim 1\text{--}5\%$ estimated observational errors), but is computable and constitutes a UQFF first-principles prediction.
3. **Cosmological implications:** The $\kappa(z)$ law predicts that UQFF gravity was weaker in the early universe by the factor $1/(1+z)$, offering a potential explanation for the delayed gravitational collapse epoch without invoking modified gravity theories.
4. **Dual outer multiplier uniqueness:** Sombrero is the first UQFF module to employ *two* outer multipliers simultaneously ($\kappa_{\text{recession}} \times \sigma_{\text{SC}}$), whose combined effect is:

$$\kappa_{\text{recession}} \times \sigma_{\text{SC}} = 0.99374 \times (1 - 10^{-20}) \approx 0.99374$$

7. References

- PAPER_273: UQFF Gravitational Blueshift Amplifier for $z < 0$ (Andromeda Galaxy)
- PAPER_276: Friedmann $H(z)$ Expansion Coupling in UQFF (Andromeda)
- SOMBRERO_UQFF_MODULE.cpp (UQFF 2.0, Session 77)
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- Ford, H. C. et al. (1996). *ApJ*, 458, 132. (Sombrero BH mass from gas kinematics)
- Emsellem, E. et al. (2004). *MNRAS*, 352, 721. (M104 stellar kinematics and redshift)

UQFF 2.0 — All physics is additive. The $\kappa_{\text{recession}}$ factor does not replace any prior term — it is a cosmological outer-multiplier consistent with vacuum field energy propagation theory. — Daniel T. Murphy, Session 77, March 2026.